

visualization Tool for Electric Vehicle Charge and Range Analysis
using Tableau

TEAM ID	SWTID-2026-6767
PROJECT TITLE	visualization Tool for Electric Vehicle Charge and Range Analysis using Tableau
MAX MARKS	3 MARKS

6.2 – Data Connectivity

Step 1: Download and Install Required Software

Install the following applications:

1. [XAMPP](#)
2. [MySQL Workbench](#)
3. [Tableau Desktop](#)
4. [MySQL Connector/ODBC](#)

Step 2: Start MySQL Server

1. Open XAMPP Control Panel.
2. Start the **MySQL** service.
3. Verify that MySQL is running successfully.

Step 3: Create Database Schema

1. Open MySQL Workbench.
2. Connect to the local MySQL server.
3. Execute the following SQL command:

```
CREATEDATABASEelectric_vehicle;  
USE electric_vehicle
```

Step 4: Import Excel Datasets into MySQL

Import the following Excel files into tables within the electric_vehicle database:

1. Cheapestelectriccars-EVDatabase.xls
2. electric_vehicle_charging_station_list.xls

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3. ElectricCarData_Clean.xls
4. EVIndia.xls

Import Process

1. Convert .xls files to .csv if necessary.
2. In MySQL Workbench:
 - o Right-click the schema electric_vehicle.
 - o Select **Table Data Import Wizard**.
 - o Choose the dataset file.
 - o Define table name and column data types.
 - o Complete the import.

Repeat for all four datasets.

Step 5: Install MySQL ODBC Connector

1. Download and install the MySQL Connector/ODBC.
2. Ensure the connector version matches your Tableau Desktop architecture (64-bit recommended).

Step 6: Connect MySQL Database to Tableau Desktop

1. Open Tableau Desktop.
2. Under **Connect**, select **MySQL**.
3. Enter the following details:

Field	Value
Server	localhost
Database	electric_vehicle
Username	root
Password	(your MySQL password)

4. Click **Sign in** or **Connect**.

Step 7: Load Tables into Tableau

1. Select the imported tables from the electric_vehicle database.
2. Drag the required tables to the data source canvas.
3. Create relationships or joins between tables as needed.
4. Verify that the data preview loads correctly.

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My SQL Workbench:

